

Yahya Abouelmagd

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Skills

Languages : Python, Java, C/C++, Kotlin, SwiftUI, PHP, HTML/CSS, SQL, ARM64 Assembly, Verilog
Frameworks, Tools, & DBs: NodeJS, ExpressJS, ReactJS, NumPy, PyTorch, Spring, REST, MongoDB, OracleDB, GDB, Docker, K8s

Education

University of British Columbia 09/2021 – 05/2026
Bachelor of Applied Science in Computer Engineering, Dean's list 2022 Vancouver, BC

Work Experience

UBC ReSeSS Lab 01/2026 – present
Undergraduate Researcher Machine Learning, Cybersecurity

- Designing a scalable, interpretable method-level **Android malware detector** by semantically embedding application-code and filtering non-discriminative code.

Broadcom 01/2025 – 08/2025
Software Engineer Co-op Microservices, AWS, Spring, Jenkins, Java, Docker, Gradle, Git/GitHub

- Accelerated the Layer7 Gateway's code-quality checks by building a static analysis pipeline in **Jenkins**, **significantly reducing analysis time** and **annual engineering effort**.
- Improved the Layer7 Gateway's distributed caching reliability by tightening key-handling and recovery logic, reducing cache-related errors by **15%**.

Terraforma Systems Inc. 09/2024 – 12/2024
Computer Engineer Co-op Machine-Learning, Web-Development, Python, Java, JavaScript, Git/GitHub

- Deployed a **CNN-based** material classification pipeline, **reducing triage time from hours to minutes**.

DELL 07/2024 – 08/2024
Intern Test-Driven Design, Networking, Software Development Lifecycle, Git/GitHub

- Decreased packet loss by **35%** by implementing retransmission logic, acknowledgment tracking, and flow control in a custom **TCP** protocol under simulated network congestion.

Capstone Project 09/2025-04/2026

Agentic, AI Data Scientist | *Test-Driven Design, System Design, ML, Python, REST, NodeJS, Docker, MCP, Git/GitHub*

- Designing and building an end-to-end **LLM-orchestrated & ML-powered, retail analytics platform** with a conversational interface that allows non-technical users to self-serve insights in natural language.

Academic Projects

Distributed Key-Value Storage (01/2026-03/2026) | *Distributed Systems, Java, JUnit, Git/GitHub*

- Building a **fault-tolerant distributed** key-value store with **at-most-once semantics**, ensuring correct behavior under retries, message loss, and concurrent client access.

Dormdash(Placed 3rd/20) | *Stripe, Google OAuth, Microservices, Kotlin, REST, NodeJS, Docker, AWS, MongoDB, Git/GitHub*

- An Uber-like mobile application enabling students to **request, store, retrieve belongings, receive real-time updates** and **chat** with movers.
- Implemented a **greedy** routing solution, providing movers with a suggested optimal & profitable sequence of jobs, improving earnings/hour by up to **65%**.

Video Game Relational Database Management System | *React, OracleSQL, Oracle Database System, Git/GitHub*

- Built a multi-table **RDBMS** optimized for fast look-ups and multi-way joins, enabling high-performance query execution over a large game dataset.

Harvard's OS161, Scaled | *C, GDB (Debugger), Program Analysis (Test Coverage), Git/GitHub*

- Upgraded **OS/161** with synchronization, multiprocessing, and virtual memory mechanisms, to support up to **1000** concurrent processes.